

TOKMA, Russian Federation

WMO: 301270

Lat: 58.252N

Lon: 105.856E

Elev: 1440

StdP: 13.95

Time Zone: 8.00 (E08)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-49.9	-45.2	-54.3	0.2	-49.7	-49.6	0.3	-45.2	16.2	-2.8	14.4	-0.8	0.0	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.2	83.6	63.2	80.1	62.2	76.4	60.9	65.9	77.5	64.1	76.1	62.3	73.4	4.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.8	87.1	69.7	59.7	80.6	68.2	57.6	74.8	66.7	31.5	77.2	30.1	76.2	28.8	73.5	70.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.5	13.5	11.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.0	-13.7	-8.4	10.9	26.8	41.7	56.7	60.1	55.3	41.3	22.9	2.6	-10.1
	DBStd	28.97	16.45	16.53	13.49	10.98	8.02	7.87	5.47	6.42	7.39	11.03	16.79	18.47
	HDD50	10256	1976	1635	1213	697	280	29	1	26	277	839	1421	1862
	HDD65	14989	2441	2055	1678	1147	724	259	170	303	710	1304	1871	2327
	CDD50	777	0	0	0	0	23	231	314	192	17	0	0	0
	CDD65	34	0	0	0	0	1	11	18	4	0	0	0	0
	CDH74	1238	0	0	0	0	16	544	440	228	10	0	0	0
	CDH80	285	0	0	0	0	2	127	95	61	0	0	0	0
	WSAvg	4.2	3.9	4.4	4.9	5.2	4.7	3.7	3.9	3.1	3.5	4.1	4.6	4.4
	PrecAvg	15.3	0.7	0.5	0.5	0.8	1.3	2.0	2.4	2.2	1.5	1.3	1.1	0.9
(15) Precipitation	PrecMax	18.7	1.5	0.9	1.1	1.6	2.3	3.5	4.6	3.8	2.9	2.2	1.9	1.7
	PrecMin	11.8	0.2	0.1	0.2	0.4	0.5	0.7	0.6	0.6	0.5	0.4	0.6	0.3
	PrecStd	1.7	0.3	0.2	0.2	0.3	0.4	0.6	1.1	0.7	0.5	0.5	0.3	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.3	30.4	46.9	60.2	75.3	87.3	87.0	86.1	75.4	51.7	39.4	30.3
		MCWB	27.6	27.1	37.0	44.5	53.3	63.5	66.3	64.3	57.7	41.4	37.2	28.5
	2%	DB	22.2	26.2	40.1	54.0	68.8	83.7	83.0	80.4	67.4	45.5	34.4	23.2
		MCWB	21.0	24.5	32.9	41.3	50.3	62.0	64.4	62.8	53.6	38.8	33.2	22.1
	5%	DB	17.4	21.5	34.8	49.3	64.2	80.8	79.1	75.0	61.2	40.6	30.1	20.6
		MCWB	16.2	19.4	29.7	39.0	48.6	61.7	62.9	59.7	50.7	35.9	28.5	19.6
	10%	DB	12.1	16.4	30.5	44.3	59.2	76.8	75.6	70.5	55.7	36.8	25.7	16.7
		MCWB	11.2	14.8	26.6	36.2	46.1	60.1	61.5	58.9	47.7	33.6	24.2	15.8
	0.4%	WB	26.8	27.2	38.3	45.0	56.0	66.9	68.5	66.5	58.7	43.0	38.3	28.7
		MCDB	27.4	29.3	46.3	56.5	73.1	79.4	80.6	77.8	71.6	49.5	39.5	30.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	21.1	24.3	33.4	42.1	52.5	64.7	66.5	64.6	55.4	39.4	33.1	22.2
		MCDB	22.2	26.0	39.8	52.7	63.9	79.4	77.8	75.8	65.1	44.5	34.1	23.1
	5%	WB	16.3	19.5	29.9	39.3	49.9	62.9	64.8	62.5	51.9	36.5	28.7	19.6
		MCDB	17.3	21.3	34.1	47.8	61.5	77.0	75.9	72.3	59.8	40.1	30.0	20.6
	10%	WB	11.3	14.9	26.8	36.6	47.1	61.1	62.8	60.2	48.7	33.8	24.3	15.9
		MCDB	12.2	16.4	30.6	44.2	57.8	74.4	73.3	68.3	54.7	36.4	25.7	16.9
(35) Mean Daily Temperature Range	5% DB	MDBR	20.7	26.4	28.8	23.6	23.5	28.4	25.2	26.2	23.1	17.5	18.5	18.3
		MCDBR	20.2	20.2	31.0	28.9	34.3	37.4	33.9	37.8	34.9	23.8	15.0	15.9
		MCWBR	19.7	18.2	24.4	18.0	18.7	17.9	17.3	20.3	20.6	17.1	14.0	15.0
	5% WB	MCDBR	20.2	20.3	28.7	24.9	27.8	32.1	28.1	29.7	32.4	21.5	15.0	15.5
		MCWBR	19.7	18.5	23.1	16.4	16.1	16.1	15.5	17.1	19.9	16.0	14.3	14.8
(40) Clear-Sky Solar Irradiance	taub		0.261	0.293	0.315	0.347	0.377	0.418	0.482	0.410	0.335	0.312	0.268	0.242
	taud		1.998	2.022	2.061	2.141	2.288	2.254	2.054	2.253	2.449	2.314	2.056	1.954
	Ebn at Noon		172	225	258	271	271	260	237	249	256	223	167	136
	Edh at Noon		18	29	37	40	37	40	48	36	25	22	16	13
(44) All-Sky Solar Radiation	RadAvg		202	507	976	1426	1627	1812	1654	1325	823	403	213	122
	RadStd		18	36	59	87	107	141	138	99	85	45	24	16

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page